The office of records at a university has stated that the proportion of incoming female students who major in business has increased. A sample of female students taken several years ago is compared with a sample of female students this year. Results are summarized below. Has the proportion increased significantly? Test at $\alpha = 0.10$.

Step 1:

Null hypothesis (H0): p2 = p1
Alternative hypothesis (H1): p2 > p1
Where:

- p1 = 50/250 = 0.20
- p2 = 90/300 = 0.30

Step 2:

p_pool = (x1 + x2) / (n1 + n2) = (50 + 90) / (250 + 300) = 140 / 550 ≈ 0.2545

Standard error of the difference:

- SE = √([p_pool * (1 - p_pool) * (1/n1 + 1/n2)])
- SE ≈ √([0.2545 * 0.7455 * (1/250 + 1/300)])
- SE ≈ 0.037

Z-score:

- Z = (p2 - p1) / SE ≈ (0.30 - 0.20) / 0.037 ≈ 2.70

P-value:

- For Z ≈ 2.70, the p-value for the one-sided test is approximately 0.0035, which is less than $\alpha = 0.10$.

Step 3:
Conclusion
Since the p-value is less than 0.10, we reject the null hypothesis and conclude that there is a statistically significant increase in the proportion of female students majoring in business at the 10% significance level.